Ordinary Differential Equations

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1 The Method of Variation of Parameters

Consider the nonhomogeneous second-order linear equation

$$L[y] := y'' + p(t)y' + q(t)y = g(t).$$

As before, its general solution has the form $y = y_h + y_p$, where $L[y_h] = 0$ and $L[y_p] = g$. Undetermined coefficients is convenient for polynomial, exponential, and trigonometric forcing terms, but forcing terms such as $\tan t$ or $\csc t$ do not lead to a finite trial of that kind. **Variation of parameters** constructs a particular solution using a known fundamental pair y_1, y_2 for $L[y] = 0$.

The method works on any interval where p, q, g are continuous and y_1, y_2 are linearly independent. It is important first to divide through by the coefficient of y'' if the equation is not already in the normalized form above.

1.1 Derivation of the Formula

The homogeneous solution is

$$y_h = C_1 y_1 + C_2 y_2.$$

Replace the constants by functions and seek

$$y_p = u_1(t)y_1(t) + u_2(t)y_2(t).$$

Choose the auxiliary condition

$$u_1' y_1 + u_2' y_2 = 0.$$

Then $y_p' = u_1 y_1' + u_2 y_2'$, and differentiating once more gives

$$y_p'' = u_1' y_1' + u_2' y_2' + u_1 y_1'' + u_2 y_2''.$$

Because $L[y_1] = L[y_2] = 0$, substitution into $L[y_p] = g$ leaves

$$u_1' y_1' + u_2' y_2' = g.$$

Thus the unknown derivatives satisfy the coefficient system

$$\begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} \begin{pmatrix} u_1' \\ u_2' \end{pmatrix} = \begin{pmatrix} 0 \\ g \end{pmatrix}.$$

Its determinant is the **Wronskian**

$$W = y_1 y_2' - y_1' y_2.$$

Since the solutions form a fundamental pair, $W \neq 0$ on the chosen interval. Solving the system and integrating gives

$$\boxed{u_1' = -\frac{y_2 g}{W}, \qquad u_2' = \frac{y_1 g}{W}, \qquad y_p = -y_1 \int \frac{y_2 g}{W}\, dt + y_2 \int \frac{y_1 g}{W}\, dt.}$$

Constants of integration only add homogeneous terms, so they can be set to zero when finding y_p.

1.2 Procedure

1. Normalize the equation so the coefficient of y'' is one.
2. Solve the homogeneous equation and choose independent solutions y_1, y_2.
3. Compute $W = y_1 y_2' - y_1' y_2$.
4. Integrate $u_1' = -y_2 g/W$ and $u_2' = y_1 g/W$.
5. Form $y_p = u_1 y_1 + u_2 y_2$, then add y_h.

1.3 Example: Forcing by $3\csc t$

Find the general solution of

$$y'' + 4y = 3\csc t.$$

Solution.

Work on an interval $(k\pi, (k+1)\pi)$, where $\sin t \neq 0$. The homogeneous characteristic equation $r^2 + 4 = 0$ gives

$$y_1 = \cos 2t, \qquad y_2 = \sin 2t, \qquad W = 2\cos^2 2t + 2\sin^2 2t = 2.$$

Consequently,

$$u_1' = -\frac{3\sin 2t}{2\sin t} = -3\cos t, \qquad u_1 = -3\sin t,$$

and, using $\cos 2t = 1 - 2\sin^2 t$,

$$u_2' = \frac{3\cos 2t}{2\sin t} = \frac{3}{2}\csc t - 3\sin t,$$
$$u_2 = \frac{3}{2}\ln\left|\tan\frac{t}{2}\right| + 3\cos t.$$

The particular solution simplifies via $\sin 2t\cos t - \cos 2t \sin t = \sin t$:

$$\begin{aligned} y_p &= -3\sin t\cos 2t + \left(\frac{3}{2}\ln\left|\tan\frac{t}{2}\right| + 3\cos t\right)\sin 2t \\ &= 3\sin t + \frac{3}{2}\sin 2t \ln\left|\tan\frac{t}{2}\right|. \end{aligned}$$

Therefore

$$\boxed{y = C_1\cos 2t + C_2\sin 2t + 3\sin t + \frac{3}{2}\sin 2t\ln\left|\tan\frac{t}{2}\right|.}$$

The formula is interpreted separately on each interval that does not contain a zero of $\sin t$.

1.4 Example: Forcing by $\tan t$

Find the general solution of

$$y'' + y = \tan t.$$

Solution.

Choose an interval where $\cos t \neq 0$, for example $(-\pi/2, \pi/2)$. The homogeneous equation has fundamental solutions

$$y_1 = \cos t, \qquad y_2 = \sin t, \qquad W = 1.$$

The parameter derivatives are

$$\begin{aligned} u_1' &= -\sin t \tan t = \cos t - \sec t, \\ u_2' &= \cos t \tan t = \sin t. \end{aligned}$$

Thus we may take

$$u_1 = \sin t - \ln|\sec t + \tan t|, \qquad u_2 = -\cos t.$$

The mixed sine-cosine terms cancel in $u_1 y_1 + u_2 y_2$, leaving

$$y_p = -\cos t \ln|\sec t + \tan t|.$$

Hence

$$\boxed{y = C_1 \cos t + C_2 \sin t - \cos t \ln|\sec t + \tan t|.}$$

The solution is valid on any interval that contains no pole of $\tan t$.

1.5 Key Points

- The Wronskian must be nonzero on the interval of work.
- The forcing term in the formula is the one from the normalized equation $y'' + py' + qy = g$.
- Different antiderivatives for u_1, u_2 produce particular solutions that differ only by homogeneous terms.
- The method can require harder integrals than undetermined coefficients, but it handles a much wider range of forcing terms.